MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER

JAN TYPE 1N1084

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon 250 milliampere rectifiers, in accordance with Specification MIL-S-19500, and as specified herein. This device shall not be used for new design.

1.2 Maximum ratings.

	V_r	V_R	I_o (T_A = 100°C)	I_o (T_A = 150°C)	i_f (surge) (1/120-sec; T_A = 100°C)
	v(pk)	Vdc	mAdc	mAdc	A
Maximum	480	400	750	250	30

TEMPERATURE RANGE (OPERATING): T_A = -65°C to +150°C

TEMPERATURE RANGE (NONOPERATING): T_A = -65°C to +150°C

BAROMETRIC PRESSURE REDUCED (ALTITUDE OPERATION): 8 mm Hg., min.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with Specification MIL-S-19500, and as specified herein.

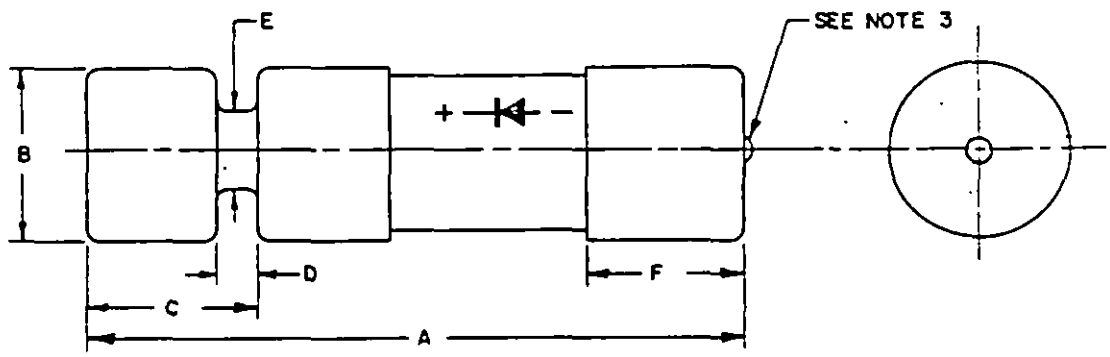
3.2 Abbreviations and symbols. The abbreviations and symbols used herein are defined in Specification MIL-S-19500.

3.3 Design and construction. The semiconductor diode shall be of the design, construction, and physical dimensions specified in figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I and II.

FSC 5960

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LTR	DIMENSIONS			
	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.975	1.035	24.77	26.28
B	.240	.260	6.10	6.60
C		.250		6.35
D	.052	.072	1.32	1.82
E		.125		3.17
F		.750		6.35

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 millimeter) are given for general information only, and are based upon 1 inch = 25.4 millimeters.
3. .015 inch solder blob allowance on negative end.
4. All tolerances are to be ±.01 inch unless otherwise specified.

Figure 1. Semiconductor device, diode, type 1N1084.

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Table I. Group A inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			5				
Visual and mechanical	2071			---	---	---	---
Subgroup 2			5				
Forward voltage (see 4.6.1)	4011	I_F = 3 Adc (pulsed)		V_F	---	1	Vdc
Reverse current	4016	V_R = 400 Vdc		I_R	---	5	uAdc
Reverse current at peak reverse voltage	4016	v_r = 480 v(pk)		I_r	---	100	uAdc
Subgroup 3			5				
Reverse current	4016	T_A = 150°C; V_R = 400 Vdc		I_R	---	500	uAdc
Reverse current, average (see 4.6.3)	4046	T_A = 150°C; I_O = 250 mAdc; v_r = 400 v (pk)		I_{RO}	---	200	uAdc

Table II. Group B inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Physical dimensions	2066			---	---	---	---
Subgroup 2			15				
Temperature cycling	1051	Test cond. C; T(high) = 150° ± 5° C		---	---	---	---
Thermal shock	1056	Test cond. A		---	---	---	---
Moisture resistance	1021	Omit initial conditioning		---	---	---	---
End points:							
Forward voltage (see 4.6.1)	4011	I_F = 3 Adc (pulsed)		V_F	---	1.0	Vdc
Reverse current	4016	V_R = 400 Vdc		I_R	---	5.0	uAdc
Subgroup 3			15				
Shock	2016	Nonoperating; 500 G, t = 1 msec; 5 blows each in orientations: X_1, Y_1 (total 10 blows)		---	---	---	---
Vibration, variable frequency	2056	**Operating; orientations X_1, Y_1, Monitor during test; Change in reverse current.**		---	---	---	---
Constant acceleration	2006	Nonoperating; 10,000 G; orientations X_1, Y_1		---	---	---	---
End points:							
(Same as subgroup 2)							

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Table II. Group B inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 6			15				
Barometric pressure reduced (altitude operation)	1001	V_R = 400 Vdc Pressure = 8 mm Hg.		---	---	---	---
End points:							
(Same as subgroup 2)							
Subgroup 5			15				
Salt atmosphere (corrosion) (see 4.6.2)	1041			---	---	---	---
End points:							
(Same as subgroup 2)							
Subgroup 6			15				
Surge current	4066	T_A = 100°C; Tan, 1/120 sec/surges; 1 surge /minute; i_f (surge) = 30a; V_{RM}(wkg) = 400 v(pk); I_o = 750 mAdc		---	---	---	---
End points:							
(Same as subgroup 2)							
Subgroup 7			15				
Reverse current at peak reverse voltage	4016	T_A = -65°C; v_R = 480 v (pk)		I_r	---	100	uAdc
End points:							
(Same as subgroup 2)							
Subgroup 8			λ = 10				
Steady state operation life	1026	T_A = 100 ± 3°C V_{RM}(wkg) = 400 v (pk) I_o = 750 mAdc		---	---	---	---
End points:							
Forward voltage (see 4.6.1)	4011	I_F = 3 Adc (pulsed)		V_F	---	1.1	Vdc
Reverse current	4016	V_R = 400 Vdc		I_R	---	10.0	uAdc
Subgroup 9			λ = 10				
High-temperature life (nonoperating)	1031	T_A = 150° ± 3°C		---	---	---	---
End points:							
(Same as subgroup 8)							

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3.5 Marking. The marking shall be placed on each device in accordance with MIL-STD-19500. The polarity shall be indicated by graphic symbol with the arrow pointing in the direction of easier current flow.

3.5.1 The manufacturer, at his option, may omit the following markings:

(a) Manufacturer's identification.
(b) Country of origin.

3.6 Salt atmosphere (corrosion). The device shall be examined for evidence of corrosion and legibility of marking before the specified measurements are made (see 4.6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with Specification MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I and II.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A and B inspections.

4.4 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.5 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II. These tests shall be performed for qualification and every six months thereafter.

4.6 Methods of examination and test. Methods of examination and test shall be as specified in tables I and II, and as follows:

4.6.1 Pulse technique. Pulse measurements shall be in accordance with section 4 of MIL-STD-750.

4.6.2 Salt atmosphere (corrosion). The markings shall be legible after the test. There shall be no evidence (when examined with no magnification) of flaking or pitting of the finish or corrosion that will interfere with the mechanical and electrical application of the devices.

4.6.3 Duty cycle requirements. This test shall be conducted with a half-sine voltage wave of the specified peak voltage impressed across the diode in the reverse direction, followed by a half-sine current wave of the specified rectified current value. The forward conduction angle of the rectified current shall not be greater than 180° nor less than 130°; and the power shall be equal to or greater than that of a half-sine wave.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with Specification MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in Specification MIL-S-19500 are applicable to this specification.

Custodians:
Army - EL
Navy - Ships
Air Force - 11

Review activities:
Army - EL, MU
Navy - SH
Air Force - 11, 17, 85

User activities:
Army - EL, MI
Navy - AR, MC, CG

Preparing activity:
Army - EL

Project No. 5960-1980

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